"FORMULATION OF NOURISHING HAIR DYE USING Areca catechu .L PEDUNCLE & HUSK EXTRACT FOR STRONG AND LUSTROUS HAIR"

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ABSTRACT

Natural hair dye formulations using areca husk and saw palmetto offer an eco-friendly alternative to conventional chemical dyes. These formulations inhibit 5-alpha reductase, reducing DHT production, which helps prevent androgenic alopecia. By promoting hair regrowth and strengthening strands, they combine therapeutic benefits with safe, sustainable hair care.

INTRODUCTION

Areca husk, a widely available agricultural by-product, has emerged as a promising source of bioactive compounds such as catechin and epigallocatechin. These tannins are known for their potent 5-alpha reductase inhibitory properties, effectively blocking dihydrotestosterone (DHT) synthesis a key factor in male pattern baldness. This study presents a novel approach to detect and quantify the amount of tannins in areca husk using UV spectrometry, providing a precise assessment of its bioactive potential. Additionally, the bioactive extract is formulated into a natural herbal hair dye that not only imparts color but also addresses the underlying cause of hair loss.

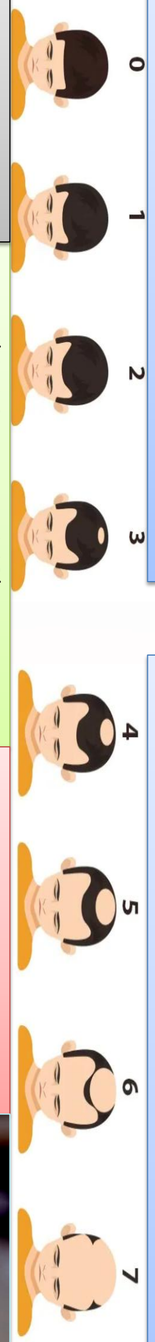
OBJECTIVES:-

- To Extract Phytochemical components from Areca nut Peduncle and Husk.
- To Perform Biochemical analysis for presence of Tannin.
- To Formulate and characterize the Hair Dye

PROBLEM STATEMENT

Chemical hair dyes often lead to irritation and premature hair graying. The market lacks natural alternatives that combat graying & promotes hair re-growth. For those who want to maintain natural black colour without harsh chemicals.

NORWOOD SCALE



METHODOLOGY

RAW MATERIAL SELECTION AND IDENTIFICATION
↓
EXTRACTION USING SOXHALET APPARATUS
↓
PHYTOCHEMICAL ANALYSIS FOR TANNIN
↓
OIL INFUSION
↓
FORMULATION OF HAIR DYE
↓
BIOCHEMICAL ANALYSIS OF HAIR DYE
↓
PRODUCT

RESULT

Standard Tannic Acid Absorbance

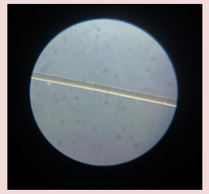
Sample Absorbance

$$\% \text{ OF TANNIN PRESENT} = \frac{\text{SAMPLE ABSORBANCE}}{\text{STANDARD ABSORBANCE}} \times \frac{\text{STANDARD CONCENTRATION}}{\text{SAMPLE CONCENTRATION}} \times \text{PURITY OF STANDARD}$$

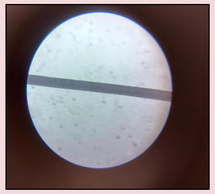
STANDARD CONCENTRATION AFTER DILUTION= 0.005
SAMPLE CONCENTRATION = 0.2
STANDARD ABSORBANCE = 0.501
SAMPLE ABSORBANCE = 0.475

$$\% \text{ OF TANNIN PRESENT} = \frac{0.475}{0.501} \times \frac{0.005}{0.2} \times 98 = 2.32\%$$

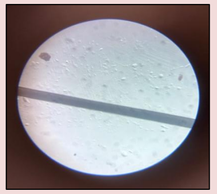
COLOUR SHADE OF DIFFERENT FORMULA



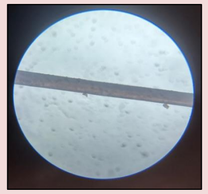
CONTROL



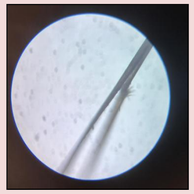
HENNA(water) + INDIGO



HENNA(Tea) + INDIGO



SEMI-SYNTHETIC

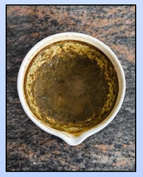


SYNTHETIC

FIGURES:

EXTRACTION AND PHYTOCHEMICAL ANALYSIS:-

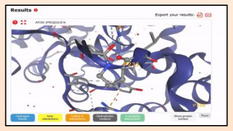
RAW MATERIAL COLLECTION



SOXHELET EXTRACT

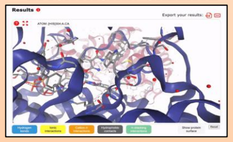
QUALITATIVE AND QUANTITATIVE ANALYSIS OF TANNIN

MOLECULAR DOCKING



CATACHIN

EPICATACHIN GALLATE



EPICATACHIN

OIL INFUSION

ARECAHUSK AND PEDUNCLE INFUSED INTO SAW PALMETTO AND COCONUT OIL

FORMULATION

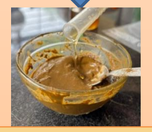
1 | 2 | 3

HENNA SOACKED IN WATER

HENNA SOACKED IN TEA DECOCTION

HENNA AND INDIGO MIXED + PPD 5%

ADDING AMLA & BRINGRAJ POWDER WITH ALOE VERA GEL



ADDITION OF OIL TO PASTE

FORMULATED HENNA PASTE

FORMULATED HENNA SOAKED IN TEA

SEMI-SYNTHETIC DYE

CONCLUSION

The synthetic hair dye caused significant damage to the hair cortex, resulting in brittleness and compromised hair health. In contrast, the semi-synthetic and natural herbal formulations, enriched with catechin and epicatechin from areca husk and saw palmetto, effectively blocked DHT and inhibited the 5-alpha-reductase enzyme, making it smooth and lustrous, proving them to be safer and superior alternatives.

REFERENCE

Raju A, De .S, Ray M.K, Degani M.S. Antituberculosis activity of polyphenols of Areca catechu. *International Journal of Mycobacteriology* 2021;10:13-8

Chougule .K, Chougule.m et al, Indian herbs that act as 5-alpha reductase inhibitors . *International journal of pharmaceutical, chemical and biological sciences,* 2017, 7(3), 265-273.

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